Name: _____ **Period:** _____ **Date:** _____

Topic 2.3 Guided Notes: Introduction to Memory

Unit 2: Cognition | AP Psychology

Directions: Read through the topic overview page below and fill in each blank as you go. Use the tables and activities to check your understanding, then head to the quiz link at the end to test yourself.

Source page: appsychreviewhub.com/overviews/topic2.3

Word Bank

Retrieval	*Phonemic Encoding*	*Explicit Memory*	*Short-Term Memory*
Long-Term Memory	*Long-Term Potentiation (LTP)*	*Information Processing Model*	*Episodic Memory*
Sensory Memory	*Semantic Encoding*	*Prospective Memory*	*Effortful Processing*
Iconic Memory	*Echoic Memory*	*Phonological Loop*	*Storage*
Central Executive	*Automatic Processing*	*Encoding*	*Semantic Memory*
Procedural Memory	*Implicit Memory*	*Working Memory*	*Structural Encoding*
Visuospatial Sketchpad			

The Big Picture: More Than Just a Video Camera

We often think of human memory like a video camera recording perfect footage of our lives. But memory is highly selective, constantly changing, and constructed in three primary stages: getting the information in (_____), holding onto it over time (_____), and getting it back out when you need it (_____). This three-stage framework, often compared to how a computer handles data, is known as the _____.

1. The Multi-Store Model: The Journey of a Memory

For a memory to become permanent, it must successfully pass through three distinct systems:

- _____: the incredibly brief initial recording of sensory information. Visual information is held as an _____ for just a fraction of a second, while auditory information is held as an _____ for a few seconds.
- _____: if you pay attention to sensory information, it moves here. It has a very limited capacity (classically thought to be about 7 items) and holds information only briefly before it is processed further or lost.
- _____: through rehearsal and encoding, information is eventually moved to this permanent, limitless storehouse.

Apply It: Which Memory Store?

Using the word bank above, identify which concept best matches each scenario.

#	Scenario	Concept
1	A camera flash leaves a brief, fading afterimage of light in a person's vision for a split second after it goes off.	
2	A woman can only hold about 6-7 random digits in her head at once when trying to memorize a new phone number.	
3	Decades later, a man can still recite every fact he learned in his high school history class.	

2. The Active Workspace: The Working Memory Model

Modern psychologists realized "short-term memory" wasn't just a passive waiting room — it's an active, scratchpad-like system now called _____. It's managed by a "boss" called the _____, which directs your attention and manages cognitive tasks. It uses sub-systems like the _____ (which temporarily stores and manipulates visual maps and shapes in your mind's eye) and the _____ (which handles auditory rehearsal, like repeating a phone number to yourself).

Apply It: Which Working Memory Component?

#	Scenario	Concept
1	Before crossing a room to grab a specific book, a man pictures the exact layout of furniture he'll need to walk around in his mind.	
2	While cooking, a chef consciously decides which task to prioritize first: stirring the sauce or checking the oven.	

📺 Video Notes: How We Make Memories (Crash Course Psychology #13)

This video also covers recall, recognition, and relearning near the start (about 1:44–2:50) and mnemonics and chunking (about 6:59–7:28). Those belong to later topics (2.6 and 2.4), so you can skip them for now. Answer the questions below on the parts that match today's material.

- What three stages does the video say Atkinson and Shiffrin used to break down memory formation, and in what order does information move through the three memory stores they described?

- According to the video, how long does information stay in short-term memory without rehearsal, how many bits of information can it hold, and what happens to it after that?

- What is the difference between explicit and implicit processes of storing information, according to the video, and what example does it give of each?

- What does Clive Wearing's case show about different types of long-term memory?

- What is the difference between shallow and deep processing, according to the video, and how does it use the name Leonidas to illustrate each?

3. How We Encode: Levels of Processing

Some information, like what you ate for lunch, skips the line and goes straight into memory without you trying — this is _____. But studying for an exam requires _____ — deliberate attention and conscious work.

The Levels of Processing Model suggests that memory depends on the depth of your encoding:

- _____: focuses purely on what a word looks like (e.g., is it in all capital letters?). _____: focuses on how a word sounds (e.g., does it rhyme with "cat"?). These shallow methods result in weak, easily forgotten memories.
- _____: requires you to think about what the word actually means and how it connects to your own life. This creates the strongest, most durable memories.

Apply It: Which Encoding Type?

#	Scenario	Concept
1	A student notices that the vocabulary word "jubilant" is written in italics on his flashcard.	
2	A student notices that the word "flair" sounds exactly like the word "flare."	
3	Without even trying, a woman remembers the song that was playing in the grocery store an hour after leaving.	

4. The Library of the Brain: Types of Long-Term Memory

_____ (Declarative) memories are ones you can consciously recall and easily describe to others.

- _____: the storybook of your life — memories of personal experiences and specific events (e.g., your 10th birthday party).
- _____: the encyclopedia of your brain — general facts, knowledge, and definitions (e.g., knowing that the capital of France is Paris).

_____ (Non-Declarative) memories operate unconsciously and are hard to explain in words. A major type is _____, the memory of how to perform physical tasks and skills (like riding a bike). Your body just "knows" how to do it.

Finally, _____ isn't about remembering the past at all — it's remembering to do something in the future (e.g., remembering to take your vitamins tonight).

Apply It: Which Type of Long-Term Memory?

#	Scenario	Concept
1	A man knows, without having to think hard about it, that Washington D.C. is the capital of the United States.	
2	A woman can tie her shoelaces perfectly without consciously thinking through each individual step.	

5. The Biology of Memory: Strengthening the Connections

Memories are stored in complex webs of neurons communicating with each other. When you learn something new and practice it repeatedly, the synaptic connections between those neurons become stronger and more efficient at firing. This biological process is called _____. As neuroscientists like to say: "Neurons that fire together, wire together."

6. Don't Trip Up! (Common Misconceptions)

_____ simply refers to the brief storage capacity. _____ refers to the active manipulation of that information to solve a problem.

_____ sounds like "Episode" (as in, an episode of a TV show starring YOU as the main character).
_____ means "meaning" (as in, the factual meaning of words and concepts).

7. Quick Check: Review Questions

Circle the best answer for each question below.

1. While distracted and staring at his phone, a man is asked by his friend, "Are you even listening to me?" Despite not consciously paying attention, he is able to accurately repeat his friend's last sentence word for word because the sound briefly lingered in his memory. This type of sensory memory is called:

- **A.** Iconic Memory
- **B.** Echoic Memory
- **C.** Working Memory
- **D.** Episodic Memory

2. To remember a new phone number just long enough to dial it, a man repeats the digits to himself over and over in his head. Which component of working memory is he relying on?

- **A.** Central Executive
- **B.** Visuospatial Sketchpad
- **C.** Echoic Memory
- **D.** Phonological Loop

3. A student effortlessly remembers exactly what she ate for breakfast that morning and the specific route she walked to get to school, despite never having deliberately tried to memorize either detail. This is best described as:

- **A.** Semantic Encoding
- **B.** Structural Encoding
- **C.** Automatic Processing
- **D.** Effortful Processing

4. While studying for her AP Psychology exam, a student has to deliberately focus, pay close attention, and consciously work to memorize dense vocabulary terms. This type of deliberate, conscious encoding is called:

- **A.** Sensory Memory
- **B.** Procedural Memory
- **C.** Automatic Processing
- **D.** Effortful Processing

5. Rather than simply memorizing a vocabulary word's spelling or sound, a student connects the word's actual meaning to a personal memory from her own life, creating an unusually strong and durable memory for that term. This deep encoding strategy is called:

- **A.** Phonemic Encoding
- **B.** Automatic Processing
- **C.** Structural Encoding
- **D.** Semantic Encoding

6. A woman can vividly and consciously recall the specific details of her 10th birthday party, including who attended and what cake was served, and can easily describe this event to others. This type of memory is called:

- **A.** Prospective Memory
- **B.** Episodic Memory
- **C.** Procedural Memory
- **D.** Semantic Memory

7. Before going to bed, a man makes a mental note to remember to take his vitamins the following morning. This type of memory, focused on remembering to perform a future action rather than recalling the past, is called:

- **A.** Semantic Memory
- **B.** Procedural Memory
- **C.** Episodic Memory
- **D.** Prospective Memory

✅ Check Your Progress

Done with your notes? Head over to the Quiz to test what you've learned:
https://appsychreviewhub.com/quiz

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